

Project Documentation

1. Introduction

Project Title: Smart Library Request Workflow in ServiceNow

Team Members:

- Member 1 – Developer
- Member 2 – Workflow Designer
- Member 3 – Testing
- Member 4 – Documentation

2. Project Overview

Purpose: The purpose of this project is to automate the traditional library system using ServiceNow. It helps students request books online and allows librarians to manage approvals efficiently through a digital workflow.

Features:

- ☐ Online book request system
- ☐ Role-based access (Student & Librarian)
- ☐ Automated approval workflow using Flow Designer
- ☐ Real-time book status updates (Available, Issued, Returned)
- ☐ Email notifications
- ☐ Reports such as Most Borrowed Books

3. Architecture

Service Portal / UI:

- Forms for book request submission
- User-friendly interface for students and librarians
- UI Policies to control form behavior
- Dynamic field visibility and validation

ServiceNow Platform:

- Business logic implemented using Flow Designer
- Automation for approval and status updates
- Role-based access using ACLs
- Server-side scripting (if required)

ServiceNow Tables:

Custom Tables:

- **Book Table**
 - Book Name
 - Author
 - Status (Available/Issued)
- **Borrow Request Table**
 - Student Name
 - Book Reference
 - Request Status
 - Approval Status

4. Setup Instructions

Prerequisites:

- ServiceNow Developer Instance
- Basic knowledge of ServiceNow platform

Installation:

- Create custom tables (Book, Borrow Request)
- Define roles (Student, Librarian)
- Configure forms and fields
- Apply UI Policies
- Create Flow Designer workflow
- Configure ACLs for security
- Create reports and dashboards

5. ServiceNow Modules

- Tables (Book, Borrow Request)
- Forms & Views
- Flow Designer
- UI Policies
- Access Control Rules (ACLs)
- Reports & Dashboards

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6. Running the Application

- Login to ServiceNow instance
- Navigate to Service Portal / Application
- Submit book request
- Librarian approves/rejects request
- System updates status automatically

7. Workflow Documentation

Book Request Process

- Student logs in
- Selects available book
- Submits request
- Librarian reviews request
- Approval/Rejection
- Book status updated automatically

8. Authentication

- ☐ Role-based access control
- ☐ Roles:
 - Student → Can request books
 - Librarian → Can approve/reject requests
- ☐ ACLs ensure:
 - Only authorized users access data
 - Data security and integrity maintained

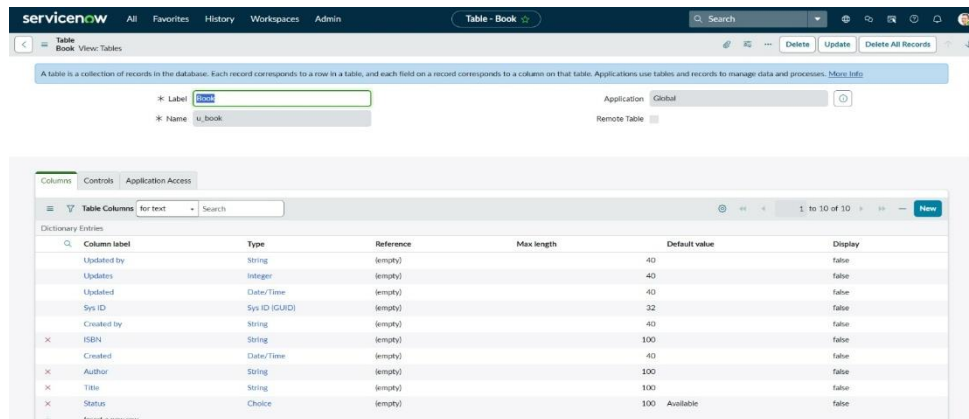
9. User Interface

- Book Request Form
- Librarian Approval Form
- Book Status View
- Reports Dashboard

10. Testing

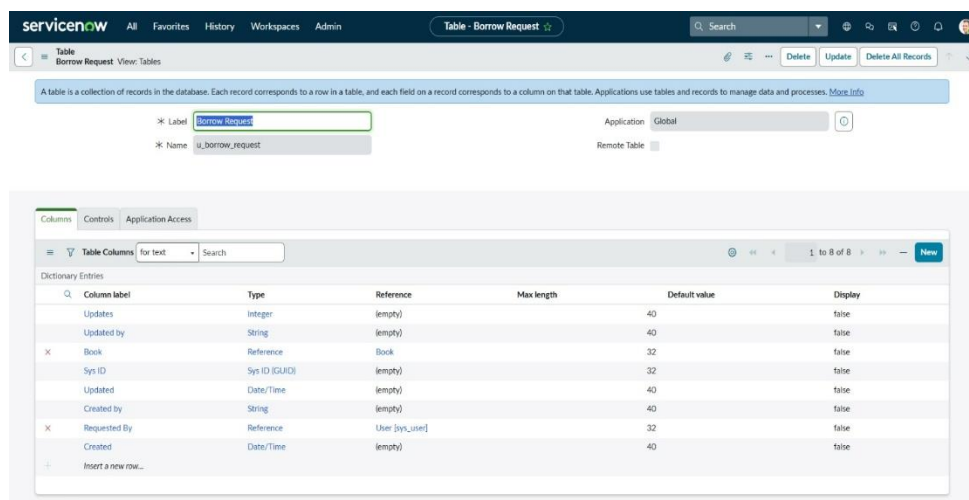
- Tested request submission
- Verified approval workflow
- Checked book status updates
- Ensured role-based access works correctly

11. Screenshots or Demo



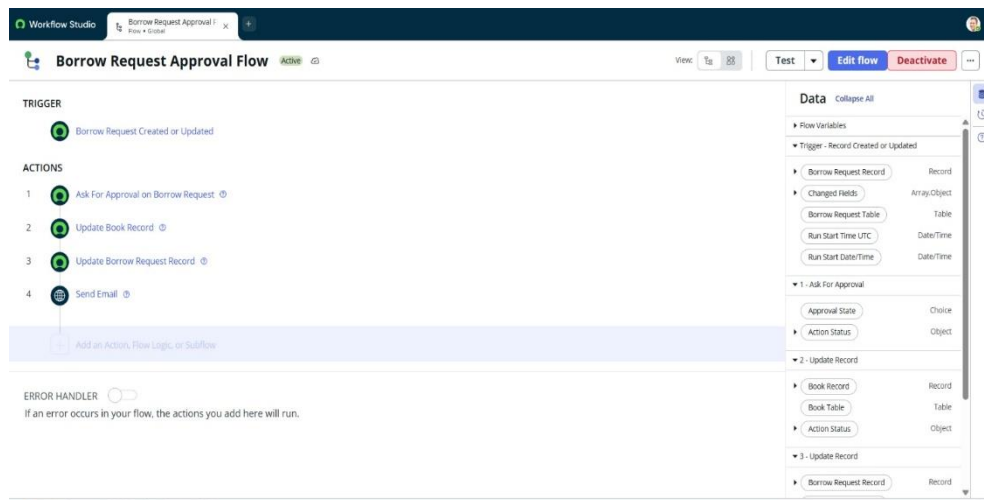
The screenshot shows the ServiceNow configuration page for the 'Table - Book'. The top navigation bar includes 'All', 'Favorites', 'History', 'Workspaces', and 'Admin'. The main header displays 'Table - Book' with a search bar and action buttons: 'Delete', 'Update', and 'Delete All Records'. Below the header, a blue banner explains that a table is a collection of records in the database. The configuration fields show 'Label' as 'Book' and 'Name' as 'u_book'. The 'Application' is set to 'Global' and 'Remote Table' is empty. The 'Columns' tab is active, showing a list of columns with their labels, types, references, max lengths, default values, and display settings. The columns are: 'Updated by' (String, empty, 40, false), 'Updates' (Integer, empty, 40, false), 'Updated' (Date/Time, empty, 40, false), 'Sys ID' (Sys ID (GUID), empty, 32, false), 'Created by' (String, empty, 40, false), 'ISBN' (String, empty, 100, false), 'Created' (Date/Time, empty, 40, false), 'Author' (String, empty, 100, false), 'Title' (String, empty, 100, false), and 'Status' (Choice, empty, 100, Available, false). A 'New' button is visible in the top right of the columns list.

Column label	Type	Reference	Max length	Default value	Display
Updated by	String	(empty)	40		false
Updates	Integer	(empty)	40		false
Updated	Date/Time	(empty)	40		false
Sys ID	Sys ID (GUID)	(empty)	32		false
Created by	String	(empty)	40		false
ISBN	String	(empty)	100		false
Created	Date/Time	(empty)	40		false
Author	String	(empty)	100		false
Title	String	(empty)	100		false
Status	Choice	(empty)	100	Available	false



The screenshot shows the ServiceNow configuration page for the 'Table - Borrow Request'. The top navigation bar includes 'All', 'Favorites', 'History', 'Workspaces', and 'Admin'. The main header displays 'Table - Borrow Request' with a search bar and action buttons: 'Delete', 'Update', and 'Delete All Records'. Below the header, a blue banner explains that a table is a collection of records in the database. The configuration fields show 'Label' as 'Borrow Request' and 'Name' as 'u_borrow_request'. The 'Application' is set to 'Global' and 'Remote Table' is empty. The 'Columns' tab is active, showing a list of columns with their labels, types, references, max lengths, default values, and display settings. The columns are: 'Updated by' (Integer, empty, 40, false), 'Updated' (String, empty, 40, false), 'Book' (Reference, Book, 32, false), 'Sys ID' (Sys ID (GUID), empty, 32, false), 'Updated' (Date/Time, empty, 40, false), 'Created by' (String, empty, 40, false), 'Requested By' (Reference, User [Sys_User], 32, false), and 'Created' (Date/Time, empty, 40, false). A 'New' button is visible in the top right of the columns list.

Column label	Type	Reference	Max length	Default value	Display
Updated by	Integer	(empty)	40		false
Updated	String	(empty)	40		false
Book	Reference	Book	32		false
Sys ID	Sys ID (GUID)	(empty)	32		false
Updated	Date/Time	(empty)	40		false
Created by	String	(empty)	40		false
Requested By	Reference	User [Sys_User]	32		false
Created	Date/Time	(empty)	40		false



12. Known Issues

- Requires internet access
- Limited customization in free instance
- Initial learning curve for ServiceNow

13. Future Enhancements

- Mobile app integration
- AI-based book recommendation
- RFID integration
- Advanced analytics dashboard.